Concept 4 | Identities and Proofs

English Student Edition • Focused formulas and guided practice

Formula opener

Proof route

$$\text{LHS} \longrightarrow \text{RHS}$$

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference

$$a^2 - b^2 = (a - b)(a + b)$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Identities and Proofs**

Q001 **Challenge**

$2x^2 + 3x + 1 = (x+1)(ax+b) + c$. **Find a,b,c.**

Solution

1. Decide that this is a polynomial identity and expand the product.
2. Match the x^2, x, and constant coefficients.
3. The constants are $a = 2$, $b = 1$, $c = 0$; substitute back.

Final answer

$a = 2,\ b = 1,\ c = 0$

Worked Solutions **Identities and Proofs**

Q002 Written

$x^2 + 6 = a(x + 1)^2 + b(x + 1) + c$

Solution

1. Expand the shifted square: the right side is ax^2+(2a+b)x+(a+b+c).
2. Match coefficients with $x^2 + 0x + 6$: $a = 1$, $2a + b = 0$, $a + b + c = 6$.
3. Solve in order to obtain $a = 1$, $b = -2$, $c = 7$.
4. Substitution reproduces the original identity.

Final answer

$a = 1,\ b = -2,\ c = 7$

Q003 Written

$x^2 - x - 3 = a(x - 1)^2 + b(x - 1) + c$

Solution

1. Expand and collect the right side as ax^2+(-2a+b)x+(a-b+c).
2. Match with $x^2 - x - 3$ and solve $a = 1$, $b = 1$, $c = -3$.
3. Check all three coefficients in the original identity.

Final answer

$a = 1,\ b = 1,\ c = -3$

Worked Solutions **Identities and Proofs**

Q004 *MCQ*

For the identity $x^2 + 6x + 5 = a(x+2)^2 + b(x+2) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2, x, and constant coefficients.
3. Read off $a = 1$, $b = 2$, $c = -3$.
4. Substitute back to check.

Correct option: C

Final answer

$(1,\ 2,\ -3)$

Q005 *MCQ*

For the identity $2x^2 + 9x + 8 = a(x+3)^2 + b(x+3) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2, x, and constant coefficients.
3. Read off $a = 2$, $b = -3$, $c = -1$.
4. Substitute back to check.

Correct option: C

Final answer

$(2,\ -3,\ -1)$

Worked Solutions **Identities and Proofs**

Q006 MCQ

For the identity $3x^2 + 28x + 65 = a(x+4)^2 + b(x+4) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2, x, and constant coefficients.
3. Read off $a = 3$, $b = 4$, $c = 1$.
4. Substitute back to check.

Correct option: D

Final answer

$(3,\ 4,\ 1)$

Q007 MCQ

For the identity $x^2 + 5x + 3 = a(x+5)^2 + b(x+5) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2, x, and constant coefficients.
3. Read off $a = 1$, $b = -5$, $c = 3$.
4. Substitute back to check.

Correct option: A

Final answer

$(1,\ -5,\ 3)$

Worked Solutions **Identities and Proofs**

Q008 Written

$a(x+1) - b(x+5) = 8x$

Solution

1. Collect coefficients: $(a-b)x + (a-5b) = 8x$.
2. Solve $a - b = 8$ and $a - 5b = 0$.
3. Hence $a = 10$ and $b = 2$; the constant term checks as zero.

Final answer

$a = 10,\ b = 2$

Q009 MCQ

For the identity $a(x+1) - b(x+3) = x - 1$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 2$, $b = 1$.
4. Check both terms.

Correct option: A

Final answer

$(2,\ 1)$

Worked Solutions **Identities and Proofs**

Q010 MCQ

For the identity $a(x+2)-b(x+4)=5x+14$, the ordered tuple (a,b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a=3$, $b=-2$.
4. Check both terms.

Correct option: B

Final answer

$(3,\ -2)$

Q011 MCQ

For the identity $a(x+3)-b(x+5)=x-3$, the ordered tuple (a,b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a=4$, $b=3$.
4. Check both terms.

Correct option: D

Final answer

$(4,\ 3)$

Worked Solutions **Identities and Proofs**

Q012 **MCQ**

For the identity $a(x+4) - b(x+6) = 9x + 44$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 5$, $b = -4$.
4. Check both terms.

Correct option: D

Final answer

$(5,\ -4)$

Q013 **Written**

$(x+a)(b+3) = 4x + 12$

Solution

1. Expand as (b+3)x+a(b+3).
2. Match $b + 3 = 4$ and $a(b+3) = 12$.
3. Therefore $b = 1$ and $a = 3$; c is not present in the printed identity and is not applicable.

Final answer

$a = 3,\ b = 1$

Q014 MCQ

For the identity $(x+a)(b+4) = 3x+6$, the ordered tuple (a, b) is:

Solution

1. Expand as (b+d)x+a(b+d).
2. Match the coefficient of x and the constant term.
3. Obtain $a = 2$, $b = -1$; no extra parameter is present.
4. Substitute back.

Correct option: B

Final answer

$(2,\ -1)$

Q015 MCQ

For the identity $(x+a)(b+5) = 5x+15$, the ordered tuple (a, b) is:

Solution

1. Expand as (b+d)x+a(b+d).
2. Match the coefficient of x and the constant term.
3. Obtain $a = 3$, $b = 0$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(3,\ 0)$

Worked Solutions **Identities and Proofs**

Q016 MCQ

For the identity $(x + a)(b + 6) = 7x + 28$, the ordered tuple (a, b) is:

Solution

1. Expand as (b+d)x+a(b+d).
2. Match the coefficient of x and the constant term.
3. Obtain $a = 4$, $b = 1$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(4,\ 1)$

Q017 Written

$2x^2 + 5x + 4 = (x + 1)(ax + b) + c$

Solution

1. Expand the right side to ax^2+(a+b)x+(b+c).
2. Match $(a, a + b, b + c) = (2, 5, 4)$.
3. Solve $a = 2$, $b = 3$, $c = 1$ and check the constant term.

Final answer

$a = 2,\ b = 3,\ c = 1$

Worked Solutions **Identities and Proofs**

Q018 **Written**

$(ax+b)(x+1)=4x^2+cx+1$

Solution

1. Expand: $ax^2+(a+b)x+b$.
2. Match the x^2 and constant coefficients to get $a=4$ and $b=1$.
3. Then $c=a+b=5$.

Final answer

$a=4,\ b=1,\ c=5$

Q019 **Written**

$2x^2-7x+8=(x-3)(ax+b)+c$

Solution

1. Expand to $ax^2+(b-3a)x+(-3b+c)$.
2. Match with (2,-7,8).
3. Solve $a=2$, $b=-1$, $c=5$ and substitute back.

Final answer

$a=2,\ b=-1,\ c=5$

Worked Solutions **Identities and Proofs**

Q020 *MCQ*

For the identity $(ax-1)(x+1)+c=x^2-3$, the ordered tuple (a,c) is:

Solution

1. Expand the product to ax^2+(ah+b)x+bh+c.
2. Match the x^2, x, and constant coefficients.
3. Use $a=1$ and $c=-2$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(1,\ -2)$

Q021 *MCQ*

For the identity $(ax+2)(x+2)+c=2x^2+6x+3$, the ordered tuple (a,c) is:

Solution

1. Expand the product to ax^2+(ah+b)x+bh+c.
2. Match the x^2, x, and constant coefficients.
3. Use $a=2$ and $c=-1$, then verify the middle coefficient.
4. Check the full identity.

Correct option: C

Final answer

$(2,\ -1)$

Worked Solutions **Identities and Proofs**

Q022 MCQ

For the identity $(ax+3)(x+3)+c=3x^2+12x+9$, the ordered tuple (a,c) is:

Solution

1. Expand the product to $ax^2+(ah+b)x+bh+c$.
2. Match the x^2, x, and constant coefficients.
3. Use $a=3$ and $c=0$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(3,\ 0)$

Q023 Written

$$\frac{3x+1}{(x+2)(x-3)}=\frac{a}{x+2}+\frac{b}{x-3}$$

Solution

1. Exclude $x=-2$ and $x=3$ from the original denominator.
2. Multiply by (x+2)(x-3): $3x+1=a(x-3)+b(x+2)$.
3. Match coefficients: $a+b=3$ and $-3a+2b=1$, giving $a=1$, $b=2$.
4. Substitute a and b back into the rational identity.

Final answer

$a=1,\ b=2$

Worked Solutions **Identities and Proofs**

Q024 **Written**

$$\frac{x+8}{(x-2)(x+3)} = \frac{a}{x-2} + \frac{b}{x+3}$$

Solution

1. Exclude $x = 2$ and $x = -3$.
2. Clear denominators: $x + 8 = a(x + 3) + b(x - 2)$.
3. Match $a + b = 1$ and $3a - 2b = 8$, giving $a = 2$, $b = -1$.
4. Substitute and verify.

Final answer

$a = 2,\ b = -1$

Q025 **Written**

$$\frac{1}{(2x-1)(x+1)} = \frac{a}{2x-1} + \frac{b}{x+1}$$

Solution

1. Exclude $x = 1/2$ and $x = -1$.
2. Clear denominators: $1 = a(x + 1) + b(2x - 1)$.
3. Match $a + 2b = 0$ and $a - b = 1$, giving $a = 2/3$, $b = -1/3$.
4. Verify in the original decomposition.

Final answer

$a = \frac{2}{3},\ b = -\frac{1}{3}$

Worked Solutions **Identities and Proofs**

Q026 Written

$$\frac{a}{x-1}+\frac{b}{x-5}=\frac{5x-1}{(x-1)(x-5)}$$

Solution

1. Exclude $x = 1$ and $x = 5$.
2. Clear denominators: $a(x-5)+b(x-1)=5x-1$.
3. Match $a+b=5$ and $-5a-b=-1$, giving $a=-1$, $b=6$.
4. Check the original rational identity.

Final answer

$a=-1,\ b=6$

Q027 Written

$$\frac{3x-8}{3x^2+5x-2}=\frac{a}{x+2}+\frac{b}{3x-1}$$

Solution

1. Factor $3x^2+5x-2=(x+2)(3x-1)$.
2. Exclude $x=-2$ and $x=1/3$, then clear denominators.
3. Match $3a+b=3$ and $-a+2b=-8$, giving $a=2$, $b=-3$.
4. Check the factored rational identity.

Final answer

$a=2,\ b=-3$

Worked Solutions **Identities and Proofs**

Q028 **Written**

$$f(t)=\frac{22t+15}{8t^2+10t+3}=\frac{a}{2t+1}+\frac{b}{4t+3},\ t>0$$

Solution

1. Factor $(8t^2+10t+3=(2t+1)(4t+3))$.
2. The stated domain is (t>0); the algebraic exclusions lie outside it.
3. Clear denominators: $(22t+15=a(4t+3)+b(2t+1))$.
4. Match $(4a+2b=22)$ and $(3a+b=15)$ to get $(a=4,\ b=3)$.

Final answer

$a=4,\ \ b=3$

Q029 **Written**

Prove that $(x+y)^2-(x-y)^2=4xy$.

Solution

1. Expand both conjugate squares on the left.
2. Remove the parentheses: $x^2+2xy+y^2-x^2+2xy-y^2$.
3. Collect like terms to obtain 4xy, which is the right-hand side.

Final answer

$4xy$

Worked Solutions **Identities and Proofs**

Q030 Written

Prove that $(2a+b)^2-(a+2b)^2=3\left(a^2-b^2\right)$.

Solution

1. Recognise a difference of squares: $U^2-V^2=(U-V)(U+V)$.
2. Set $U=2a+b$ and $V=a+2b$; then $U-V=a-b$ and $U+V=3a+3b$.
3. Multiply to get $(a-b)(3a+3b)=3(a^2-b^2)$.

Final answer

$3a^2-3b^2$

Q031 MCQ

For the identity $-(x-2y+1)^2+(x+2y+1)^2=2y\,(4x+4)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy+8y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A

Final answer

$8xy+8y$

Worked Solutions **Identities and Proofs**

Q032 MCQ

For the identity $-(x-3y+2)^2+(x+3y+2)^2=3y(4x+8)$, **the value of the simplified left-hand side is:**

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $12xy+24y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$12xy+24y$

Q033 MCQ

For the identity $-(x-y+3)^2+(x+y+3)^2=y(4x+12)$, **the value of the simplified left-hand side is:**

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $4xy+12y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$4xy+12y$

Worked Solutions **Identities and Proofs**

Q034 MCQ

For the identity $-(x-2y+4)^2+(x+2y+4)^2=2y(4x+16)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy+32y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A

Final answer

$8xy+32y$

Q035 Written

Prove that $(x+y)^2+(x-y)^2=2x^2+2y^2$.

Solution

1. Expand both squares on the left.
2. Add $x^2+2xy+y^2$ and $x^2-2xy+y^2$.
3. The mixed terms cancel, leaving $2x^2+2y^2$.

Final answer

$2x^2+2y^2$

Worked Solutions **Identities and Proofs**

Q036 MCQ

For the identity $(x-2y+1)^2+(x+2y+1)^2=8y^2+2(x+1)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2+4x+8y^2+2$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$2x^2+4x+8y^2+2$

Q037 MCQ

For the identity $(x-3y+2)^2+(x+3y+2)^2=18y^2+2(x+2)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2+8x+18y^2+8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$2x^2+8x+18y^2+8$

Worked Solutions **Identities and Proofs**

Q038 MCQ

For the identity $(x - y + 3)^2 + (x + y + 3)^2 = 2y^2 + 2(x + 3)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 12x + 2y^2 + 18$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$2x^2 + 12x + 2y^2 + 18$

Q039 MCQ

For the identity $(x - 2y + 4)^2 + (x + 2y + 4)^2 = 8y^2 + 2(x + 4)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 16x + 8y^2 + 32$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$2x^2 + 16x + 8y^2 + 32$

Worked Solutions **Identities and Proofs**

Q040 Written

Prove that $(a+2b)^2 - 4ab = (a-2b)^2 + 4ab$.

Solution

1. Expand the left-hand side: $a^2 + 4ab + 4b^2 - 4ab = a^2 + 4b^2$.
2. Expand the right-hand side: $a^2 - 4ab + 4b^2 + 4ab = a^2 + 4b^2$.
3. Both sides simplify to the same expression.

Final answer

$a^2 + 4b^2$

Q041 MCQ

For the identity $-(x-2y+3)^2 + (x+2y-1)^2 = (8x+8)(y-1)$, **the value of the simplified left-hand side is:**

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy - 8x + 8y - 8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$8xy - 8x + 8y - 8$

Worked Solutions **Identities and Proofs**

Q042 MCQ

For the identity $-(2x-3y+5)^2+(2x+3y-1)^2=(24x+24)(y-1)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $24xy-24x+24y-24$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$24xy-24x+24y-24$

Q043 Written

Prove that $\left(x^2+x+1\right)\left(x^2-x+1\right)=x^4+x^2+1$.

Solution

1. Group the factors as $[(x^2+1)-x][(x^2+1)+x]$.
2. Use $(U-V)(U+V)=U^2-V^2$ with $U=x^2+1$ and $V=x$.
3. Simplify $(x^2+1)^2-x^2$ to x^4+x^2+1.

Final answer

x^4+x^2+1

Worked Solutions **Identities and Proofs**

Q044 **Written**

Prove that $\left(a^2 - ab + b^2\right)\left(a^2 + ab + b^2\right) = \left(a^2 + b^2\right)^2 - (ab)^2$.

Solution

1. Group the factors as $[(a^2 + b^2) - ab][(a^2 + b^2) + ab]$.
2. Apply $(U - V)(U + V) = U^2 - V^2$ with $U = a^2 + b^2$ and $V = ab$.
3. This gives $(a^2 + b^2)^2 - (ab)^2$, exactly the right-hand side.

Final answer

$a^4 + a^2b^2 + b^4$

Q045 **Written**

Prove that $\left(a^2 + b^2\right)\left(c^2 + d^2\right) = (ac - bd)^2 + (ad + bc)^2$.

Solution

1. Expand the left-hand product to $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$.
2. Expand the first right-hand square: $a^2c^2 - 2abcd + b^2d^2$.
3. Expand the second right-hand square: $a^2d^2 + 2abcd + b^2c^2$.
4. Add the two expansions; the -2abcd and +2abcd terms cancel, leaving the left-hand expression.

Final answer

$a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Worked Solutions **Identities and Proofs**

Q046 **Written**

Prove that $\left(x^2+1\right)\left(y^2+1\right)=(xy+1)^2+(x-y)^2$.

Solution

1. Expand the left-hand side to $x^2y^2+x^2+y^2+1$.
2. Expand the right-hand side: $x^2y^2+2xy+1+x^2-2xy+y^2$.
3. The +2xy and -2xy terms cancel, leaving the same expression as the left.

Final answer

$x^2y^2+x^2+y^2+1$

Q047 **Written**

Prove that $\left(a^2-b^2\right)\left(c^2-d^2\right)=(ac-bd)^2-(ad-bc)^2$.

Solution

1. Expand the first right-hand square: $a^2c^2-2abcd+b^2d^2$.
2. Expand the second right-hand square: $a^2d^2-2abcd+b^2c^2$.
3. Subtract; the mixed terms cancel, giving $a^2c^2-a^2d^2-b^2c^2+b^2d^2$.
4. Factor the result as $(a^2-b^2)(c^2-d^2)$.

Final answer

$a^2c^2-a^2d^2-b^2c^2+b^2d^2$

Worked Solutions **Identities and Proofs**

Q048 Written

Prove that $(a+b)^3 - 3ab\,(a+b) = a^3 + b^3$.

Solution

1. Expand $(a+b)^3$ to $a^3 + 3a^2b + 3ab^2 + b^3$.
2. Expand 3ab(a+b) to $3a^2b + 3ab^2$.
3. Subtract the matching cross terms; the result is $a^3 + b^3$.

Final answer

$a^3 + b^3$

Q049 Written

Prove that $(a+b+c)\left(a^2 + b^2 + c^2 - ab - bc - ca\right) = a^3 + b^3 + c^3 - 3abc$.

Solution

1. Distribute a+b+c across the second bracket.
2. Write all products in a fixed order: a^3, b^3, c^3, then the mixed terms.
3. Each mixed term cancels with its opposite, while -abc appears three times.
4. The remaining expression is $a^3 + b^3 + c^3 - 3abc$.

Final answer

$a^3 - 3abc + b^3 + c^3$

Worked Solutions **Identities and Proofs**

Q050 **Challenge**

For the integrated identity $\left(x^2+1\right)\left(-(x-2y+1)^2+(x+2y+1)^2\right)=2y\,(4x+4)\left(x^2+1\right)$, the value of the simplified left-hand side is:

Solution

1. Recognise the cubic or symmetric pattern before expanding.
2. Expand in descending degree and collect like terms.
3. The cross terms cancel, leaving $8x^3y+8x^2y+8xy+8y$.
4. Therefore the proposed equality is proved.

Correct option: B

Final answer

$8x^3y+8x^2y+8xy+8y$

Worked Solutions **Identities and Proofs**

Q051 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

Solution

1. Use $x + y = 1$, so $(x + y)^2 = 1$.
2. Expand: $x^2 + 2xy + y^2 = 1$, hence $x^2 + y^2 + 1 = 2 - 2xy$.
3. Since $x + y = 1$, $2 - 2xy = 2(x + y - xy)$. Therefore the equality holds.

Final answer

$x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.

Solution

1. Square the condition: $(x + y)^2 = 1$.
2. Expand to $x^2 + 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 1 - 2xy$.

Final answer

$x^2 + y^2 = 1 - 2xy$

Worked Solutions **Identities and Proofs**

Q053 MCQ

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 2$.
2. $y = 2 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$-2xy + 4$

Q054 MCQ

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 3$.
2. $y = 3 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$-2xy + 9$

Worked Solutions **Identities and Proofs**

Q055 MCQ

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = -1$.
2. $y = -1 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$-2xy + 1$

Q056 MCQ

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 4$.
2. $y = 4 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 16$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: A

Final answer

$-2xy + 16$

Worked Solutions **Identities and Proofs**

Q057 **Written**

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

Solution

1. Square the condition: $(x - y)^2 = 1$.
2. Expand to $x^2 - 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 2xy + 1$.

Final answer

$x^2 + y^2 = 2xy + 1$

Q058 **MCQ**

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 1$.
2. $x = y + 1$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B

Final answer

$2xy + 1$

Worked Solutions **Identities and Proofs**

Q059 *MCQ*

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 2$.
2. $x = y + 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B

Final answer

$2xy + 4$

Q060 *MCQ*

Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = -2$.
2. $x = y - 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$2xy + 4$

Worked Solutions **Identities and Proofs**

Q061 MCQ

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 3$.
2. $x = y + 3$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$2xy + 9$

Q062 Written

Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.

Solution

1. Move the right-hand side to the left: $x^2 - y^2 - x + y$.
2. Factor: $x^2 - y^2 - x + y = (x - y)(x + y - 1)$.
3. The condition makes $x + y - 1 = 0$, so the difference is zero.

Final answer

$x^2 - x = y^2 - y$

Worked Solutions **Identities and Proofs**

Q063 Written

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Solution

1. From $a + b + c = 0$, write $c = -a - b$.
2. Substitute into the left-hand side: $a^2 + b^2 + (-a - b)^2 = 2a^2 + 2ab + 2b^2$.
3. Substitute into the right-hand side: $-2[ab + b(-a - b) + (-a - b)a] = 2a^2 + 2ab + 2b^2$.
4. Both sides reduce to the same expression, so the equality is proved.

Final answer

$a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 Written

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

Solution

1. Use the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.
2. The condition gives $a + b + c = 0$, so the product on the right is zero.
3. Therefore $a^3 + b^3 + c^3 - 3abc = 0$, which proves $a^3 + b^3 + c^3 = 3abc$.

Final answer

$a^3 + b^3 + c^3 = 3abc$

Worked Solutions **Identities and Proofs**

Q065 Written

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

Solution

1. Subtract the right-hand side: $a^2 + ca - b^2 - bc$.
2. Factor by grouping: $(a - b)(a + b) + (a - b)c = (a - b)(a + b + c)$.
3. Since $a + b + c = 0$, the difference is zero; hence the two sides are equal.

Final answer

$a^2 + ca = b^2 + bc$

Q066 Written

Under the condition $a + b + c = 0$, prove that $(a + b)(b + c)(c + a) + abc = 0$.

Solution

1. From $a + b + c = 0$, write $a + b = -c$, $b + c = -a$, and $c + a = -b$.
2. Then $(a + b)(b + c)(c + a) = (-c)(-a)(-b) = -abc$.
3. Adding abc gives zero, so the required equality is proved.

Final answer

$(a + b)(b + c)(c + a) + abc = 0$

Worked Solutions **Identities and Proofs**

Q067 **Written**

Under the condition $a + b + c = 0, \quad abc \neq 0$**, prove that** $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$**.**

Solution

1. Because $a + b + c = 0$, $b + c = -a$, $c + a = -b$, and $a + b = -c$.
2. The non-zero condition ensures that all three denominators are non-zero.
3. Therefore each fraction equals -1: $a/(b + c) = b/(c + a) = c/(a + b) = -1$.
4. Their sum is $-1 - 1 - 1 = -3$.

Final answer

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$$

Worked Solutions **Identities and Proofs**

Q068 Challenge

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

Solution

1. Use $x + y = 3$ to replace each cyclic denominator.
2. The non-zero condition keeps every original denominator defined.
3. Each term simplifies and the total becomes $-2xy + 10$.
4. Thus the required constant sum is proved.

Correct option: C

Final answer

$-2xy + 10$

Worked Solutions **Identities and Proofs**

Q069 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$.

Solution

1. Let $a/b = c/d = k$, so $a = bk$ and $c = dk$.
2. Substitute into the left-hand side: $ab/(a^2+b^2)=k/(k^2+1)$.
3. Substitute into the right-hand side: $cd/(c^2+d^2)=k/(k^2+1)$.
4. The two sides are equal.

Final answer

$$\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$$

Q070 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2+d^2}$$

Worked Solutions **Identities and Proofs**

Q071 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Q072 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions **Identities and Proofs**

Q073 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions **Identities and Proofs**

Q074 **Challenge**

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions **Identities and Proofs**

Q075 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions **Identities and Proofs**

Q076 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

Solution

1. Let $a = bk$ and $c = dk$.
2. The left-hand side becomes $k(b^2 + d^2)/(b^2 - d^2)$.
3. The right-hand side becomes $k^2(b^2 + d^2)/[k^2(b^2 - d^2)]$, which is the same expression.
4. Hence the equality is proved.

Final answer

$$\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$$

Worked Solutions **Identities and Proofs**

Q077 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions **Identities and Proofs**

Q078 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions **Identities and Proofs**

Q079 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions **Identities and Proofs**

Q080 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions **Identities and Proofs**

Q081 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Solution

1. Write $a = bk$ and $c = dk$.
2. Then $(a+b)/(a-b) = (bk+b)/(bk-b) = (k+1)/(k-1)$.
3. Similarly $(c+d)/(c-d) = (dk+d)/(dk-d) = (k+1)/(k-1)$.
4. The equality follows.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q082 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions **Identities and Proofs**

Q083 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Q084 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions **Identities and Proofs**

Q085 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{c+d}{c-d}$$

Q086 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

Solution

1. Substitute $a = bk$ and $c = dk$.
2. The left-hand side becomes (3k+ $k)/(3+1) = k$.
3. The right-hand side becomes $(2k-k)/(2-1) = k$.
4. Thus both sides are equal.

Final answer

$$\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$$

Worked Solutions **Identities and Proofs**

Q087 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{2a-c}{2b-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{2a-c}{2b-d}$$

Q088 Written

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}, \quad \frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

Solution

1. Let $x = ak$ and $y = bk$.
2. The left-hand side is $(a^2 + a^2k^2)/(a^2 - a^2k^2) = (1 + k^2)/(1 - k^2)$.
3. The right-hand side reduces to the same value.
4. Hence the equality is proved.

Final answer

$$\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$$

Worked Solutions **Identities and Proofs**

Q089 **Written**

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Solution

1. Use the equal concentration condition to write $a = bk$ and $c = dk$.
2. The first ratio becomes (k+1)/(k-1).
3. The second ratio becomes (k+1)/(k-1).
4. The required equality follows, provided the denominators are non-zero.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q090 **Written**

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$.

Solution

1. Let the common ratio be k, so $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also xy+yz+zx=k^2(ab+bc+ca).
4. Both ratios therefore equal k^2.

Final answer

$$\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$$

Worked Solutions **Identities and Proofs**

Q091 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x+y+z}{a+b+c} = \frac{x}{a}$.

Solution

1. Let the common ratio be k.
2. Then $x + y + z = k(a + b + c)$.
3. Therefore $(x + y + z)/(a + b + c) = k = x/a$.

Final answer

$$\frac{x+y+z}{a+b+c} = \frac{x}{a}$$

Q092 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$.

Solution

1. Set $x = ak$, $y = bk$ and $z = ck$.
2. The left numerator is k(a+2b+3c), so the left ratio is k.
3. The right ratio is also k because $x + y + z = k(a + b + c)$.
4. The two ratios are equal.

Final answer

$$\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$$

Worked Solutions **Identities and Proofs**

Q093 Written

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression: $x : y : z = 2 : 3 : 5$, $\frac{2xy+yz+3zx}{x^2+y^2+z^2}$.

Solution

1. Let $x = 2k$, $y = 3k$ and $z = 5k$.
2. The numerator is $(12 + 15 + 30)k^2 = 57k^2$.
3. The denominator is $(4 + 9 + 25)k^2 = 38k^2$.
4. Therefore the value is 57/38.

Final answer
$\frac{57}{38}$

Q094 Written

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression: $a : b : c = 2 : 3 : 4$, $\frac{ab+bc+ca}{a^2+b^2+c^2}$.

Solution

1. Let $a = 2k$, $b = 3k$ and $c = 4k$.
2. The numerator is $(6 + 12 + 8)k^2 = 26k^2$.
3. The denominator is $(4 + 9 + 16)k^2 = 29k^2$.
4. Therefore the value is 26/29.

Final answer
$\frac{26}{29}$

Worked Solutions **Identities and Proofs**

Q095 **Written**

Under the condition $a : b : c = 3 : 5 : 2$, prove or find the required expression: $a : b : c = 3 : 5 : 2$, $\frac{a^2+b^2-c^2}{a^2-b^2+c^2}$.

Solution

1. Let $a = 3k$, $b = 5k$ and $c = 2k$.
2. The numerator is $(9 + 25 - 4)k^2 = 30k^2$.
3. The denominator is $(9 - 25 + 4)k^2 = -12k^2$.
4. Therefore the value is $-5/2$.

Final answer

$$-\frac{5}{2}$$

Q096 **Written**

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression: $x : y : z = 2 : -3 : 4$, $\frac{x^2+y^2+z^2}{xy+yz+zx}$.

Solution

1. Let $x = 2k$, $y = -3k$ and $z = 4k$.
2. The numerator is $(4 + 9 + 16)k^2 = 29k^2$.
3. The denominator is $(-6 - 12 + 8)k^2 = -10k^2$.
4. Therefore the value is $-29/10$.

Final answer

$$-\frac{29}{10}$$

Worked Solutions **Identities and Proofs**

Q097 Written

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq 0, \quad \frac{x^2+5y^2}{x^2+xy+y^2}$.

Solution

1. Let $x = 7k$ and $y = 3k$, where k is non-zero.
2. The numerator is $(49 + 45)k^2 = 94k^2$.
3. The denominator is $(49 + 21 + 9)k^2 = 79k^2$.
4. Therefore the value is 94/79.

Final answer

$\frac{94}{79}$

Q098 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad (a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Solution

1. Let $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also ax+by+cz=k(a^2+b^2+c^2).
4. Both sides equal $k^2(a^2 + b^2 + c^2)^2$.

Final answer

$(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Worked Solutions **Identities and Proofs**

Quiz MCQ 1 Concept Quiz · MCQ

If $x^2 + 4x + 1 = (x+1)(ax+b) + c$, then (a, b, c) is... Which of the following is correct?:

Solution

1. Expand to $ax^2 + (a+b)x + b + c$.
2. Match $a = 1,\ a + b = 4,\ b + c = 1$, so $b = 3,\ c = -2$.

Correct option: A

Final answer

$(1, 3, -2)$

Quiz MCQ 2 Concept Quiz · MCQ

For the identity $\left(x^2 - 2x + 2\right)\left(x^2 + 2x + 2\right) = -4x^2 + \left(x^2 + 2\right)^2$, the value of the simplified left-hand side is:

Solution

1. Rewrite the factors as (U-V)(U+V).
2. Apply U^2-V^2 before carrying out any long expansion.
3. Simplify to $x^4 + 4$.
4. This is exactly the stated right-hand side.

Correct option: A

Final answer

$x^4 + 4$

Worked Solutions **Identities and Proofs**

Quiz MCQ 3 **Concept Quiz · MCQ**

Under the condition $a + b + c = 0$, the expression $3a^2 + 3ac$ is equal to:

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $3b^2 + 3bc$.
4. Hence the two sides are equal under the given condition.

Correct option: A

Final answer

$3b^2 + 3bc$

Quiz MCQ 4 **Concept Quiz · MCQ**

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

Solution

1. Write the variables as the stated multiples of k.
2. Substitute into the numerator and denominator.
3. Cancel the common factor and simplify to $\frac{3}{2}$.
4. Hence the required value is obtained.

Correct option: B

Final answer

$$\frac{3}{2}$$

Worked Solutions **Identities and Proofs**

Quiz WRITTEN 5 Concept Quiz · Written

Find a, b, c if $x^2 - 3x + 5 = (x-2)(ax+b) + c$.

Solution

1. Expand to $ax^2 + (b-2a)x - 2b + c$.
2. Match $a = 1,\ b - 2 = -3,\ -2b + c = 5$.
3. Therefore $b = -1,\ c = 3$.

Final answer

$a = 1,\ b = -1,\ c = 3$

Quiz WRITTEN 6 Concept Quiz · Written

Prove that $\left(x^2 - 3x + 4\right)\left(x^2 + 3x + 4\right) = -9x^2 + \left(x^2 + 4\right)^2$.

Solution

1. Rewrite the factors as (U-V)(U+V).
2. Apply U^2-V^2 before carrying out any long expansion.
3. Simplify to $x^4 - x^2 + 16$.
4. This is exactly the stated right-hand side.

Final answer

$x^4 - x^2 + 16$

Worked Solutions **Identities and Proofs**

Quiz WRITTEN 7 Concept Quiz · Written

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $4b^2 + 4bc$.
4. Hence the two sides are equal under the given condition.

Final answer

$4b^2 + 4bc$

Quiz WRITTEN 8 Concept Quiz · Written

Under the condition $x/a = y/b = z/c$, prove that $\left(a^2 + b^2 + c^2\right)\left(x^2 + y^2 + z^2\right) = (ax + by + cz)^2$.

Solution

1. Use the common multiplier from $x/a = y/b = z/c$.
2. Substitute and collect the common factor.
3. Both sides simplify to $(ax + by + cz)^2$.
4. Thus the identity is proved.

Final answer

$(ax + by + cz)^2$

Answer key

Use the question number shown on each page.

Q001 $a = 2,\ b = 1,\ c = 0$

Q002 $a = 1,\ b = -2,\ c = 7$

Q003 $a = 1,\ b = 1,\ c = -3$

Q004 C

Q005 C

Q006 D

Q007 A

Q008 $a = 10,\ b = 2$

Q009 A

Q010 B

Q011 D

Q012 D

Q013 $a = 3,\ b = 1$

Q014 B

Q015 C

Q016 C

Q017 $a = 2,\ b = 3,\ c = 1$

Q018 $a = 4,\ b = 1,\ c = 5$

Q019 $a = 2,\ b = -1,\ c = 5$

Q020 B

Q021 C

Q022 B

Q023 $a = 1,\ b = 2$

Q024 $a = 2,\ b = -1$

Q025 $a = \frac{2}{3},\ b = -\frac{1}{3}$

Q026 $a = -1,\ b = 6$

Q027 $a = 2,\ b = -3$

Q028 $a = 4,\ b = 3$

Q029 $4xy$

Q030 $3a^2 - 3b^2$

Q031 A

Q032 D

Q033 D

Q034 A

Q035 $2x^2 + 2y^2$

Q036 C

Q037 C

Q038 C

Q039 D

Q040 $a^2 + 4b^2$

Q041 D

Q042 C

Q043 $x^4 + x^2 + 1$

Q044 $a^4 + a^2b^2 + b^4$

Q045 $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Q046 $x^2y^2 + x^2 + y^2 + 1$

Q047 $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

Q048 $a^3 + b^3$

Q049 $a^3 - 3abc + b^3 + c^3$

Q050 B

Q051 $x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 $x^2 + y^2 = 1 - 2xy$

Q053 C

Q054 D

Q055 C

Q056 A

Q057 $x^2 + y^2 = 2xy + 1$

Q058 B

Q059 B

Q060 C

Q061 D

Q062 $x^2 - x = y^2 - y$

Q063 $a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 $a^3 + b^3 + c^3 = 3abc$

Q065 $a^2 + ca = b^2 + bc$

Q066 $(a + b)(b + c)(c + a) + abc = 0$

Q067 $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$

Q068 C

Q069 $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$

Q070 C

Q071 D

Q072 C

Q073 D

Q074 D

Q075 C

Q076 $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$

Q077 A

Q078 D

Q079 D

Q080 A

Q081 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q082 B

Q083 B

Q084 A

Q085 A

Q086 $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$

Q087 A

Q088 $\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$

Q089 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q090 $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$

Q091 $\frac{x+y+z}{a+b+c} = \frac{x}{a}$

Q092 $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$

Q093 $\frac{57}{38}$

Q094 $\frac{26}{29}$

Q095 $-\frac{5}{2}$

Q096 $-\frac{29}{10}$

Q097 $\frac{94}{79}$

Q098 $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a = 1,\ b = -1,\ c = 3$

Quiz WRITTEN 6 $x^4 - x^2 + 16$

Quiz WRITTEN 7 $4b^2 + 4bc$

Quiz WRITTEN 8 $(ax + by + cz)^2$